

Gadolinium

Gd (Gd I). Strongest classified transitions below 2 μm (NIST): 21 levels, 19 lines.

Gd ⁺ ionisation limit 49601.45 cm^{−1} = 6.14980 eV (201.607 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element

|
⟨
J
|

e

r

|

J
′

⟩

|

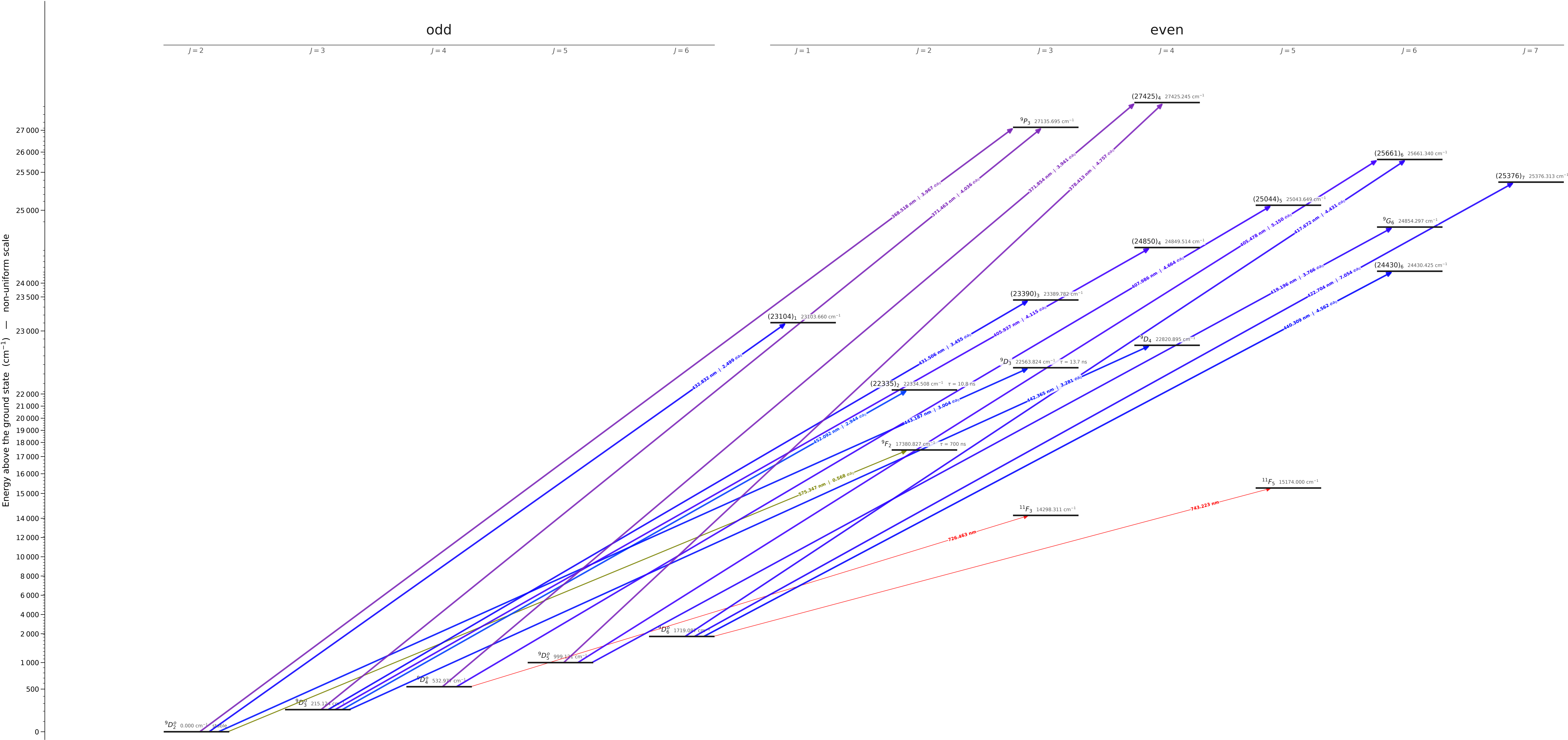
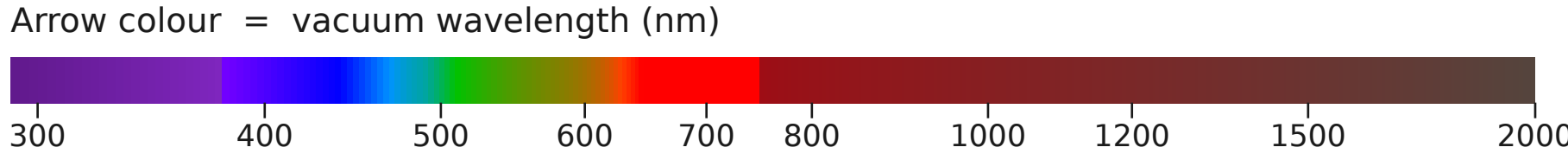
{\displaystyle |\langle J|e^{r}|J'\rangle |}

 in ea₀

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ^{−1})	Γ/2π up. (MHz)	branch (%)	Use
⁹ D ₂ ^o - ⁹ P ₃	368.5183	368.4134	813.5077	3.967	9.100e+07	-	-	proposed laser-cooling line
⁹ D ₃ ^o - ⁹ P ₃	371.4631	371.3575	807.0584	4.036	9.200e+07	-	-	371 nm repumper for the proposed 368 nm cooling scheme
⁹ D ₆ ^o - (25376) ₇	422.7038	422.5848	709.2258	7.054	8.900e+07	-	-	proposed laser-cooling line
⁹ D ₂ ^o - ⁹ D ₃	443.1873	443.0629	676.4464	3.004	3.000e+07	11.62	41.1	443.06 nm
⁹ D ₃ ^o - (22335) ₂	452.0922	451.9654	663.1224	2.944	3.800e+07	14.74	41	451.96 nm
⁹ D ₂ ^o - ⁹ F ₂	575.3466	575.1871	521.0641	0.568	6.855e+05	0.2274	48	first step
⁹ D ₄ ^o - ¹¹ F ₃	726.4626	726.2624	412.6743	-	-	-	-	726.266 nm spin-forbidden line 9D*4 - 11F3 studied as a C...
⁹ D ₆ ^o - ¹¹ F ₅	743.2229	743.0183	403.3681	-	-	-	-	743.014 nm spin-forbidden line 9D*6 - 11F5 studied as a C...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

¹⁵⁷Gd hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
⁹ D ₂ ^o	47.950	191.160	F=1/2: -48.51 F=3/2: -143.85 F=5/2: -143.45 F=7/2: +191.64
¹¹ F ₃	-77(5)	590(4)	F=3/2: +813.30 F=5/2: +179.68 F=7/2: -294.75 F=9/2: -197.32
¹¹ F ₅	84(4)	967(4)	F=7/2: -335.94 F=9/2: -587.13 F=11/2: -303.02 F=13/2: +871.08
⁹ F ₂	95.100	347.000	F=1/2: -124.33 F=3/2: -285.30 F=5/2: -264.43 F=7/2: +372.05

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁹ D ₂ ^o - ⁹ F ₂	575.347	152-156	2232.00
⁹ D ₂ ^o - ⁹ F ₂	575.347	154-156	1116.00
⁹ D ₂ ^o - ⁹ F ₂	575.347	155-156	763.00
⁹ D ₂ ^o - ⁹ F ₂	575.347	157-156	-21.00
⁹ D ₂ ^o - ⁹ F ₂	575.347	158-156	-1119.00
⁹ D ₂ ^o - ⁹ F ₂	575.347	160-156	-2233.00
⁹ D ₄ ^o - ¹¹ F ₃	726.463	158-160	964(4)
⁹ D ₄ ^o - ¹¹ F ₃	726.463	157-160	2154(20)
⁹ D ₄ ^o - ¹¹ F ₃	726.463	156-160	2563(4)
⁹ D ₄ ^o - ¹¹ F ₃	726.463	155-160	3243(12)
⁹ D ₆ ^o - ¹¹ F ₅	743.223	158-160	1054(4)
⁹ D ₆ ^o - ¹¹ F ₅	743.223	157-160	486(65)
⁹ D ₆ ^o - ¹¹ F ₅	743.223	156-160	2029(4)
⁹ D ₆ ^o - ¹¹ F ₅	743.223	155-160	2476(12)

Shift = ν(first isotope) − ν(second isotope).